Student aid refund fraud

Detection and prevention platform

Architecture Review Committee | Financial Crimes Enterprise Technology

Response to FinCEN alert

What we are detecting

Several hundred accounts over six months, three distinct fraud typologies

Ghost student

Stolen PII used to enrol fictitious students. The named recipient does not exist or never enrolled.

Identity and enrolment problem

Straw student

A real person sells their identity for compensation. KYC passes cleanly because the person is real.

Behavioural and network problem

Money mule

Refunds are rapidly moved out via cash, P2P, prepaid cards and new business accounts.

Velocity and outflow problem

3

typologies to score

< 1s

scoring latency target

0

code changes to tune logic

Assumptions I designed against

Open questions for stakeholders, and what changes in the design if the answer differs

Enforcement authority

Assumed: we may hold funds and return ACH credits.

If detect only: the core banking path disappears.

Near real time

Assumed: under one second, event driven, not inline authorisation.

If sub-100ms inline: identity calls must be pre-materialised.

Labelled outcomes

Assumed: sparse labels with 90 to 180 day confirmation lag.

If labels are rich: supervised models lead instead of rules.

Enrolment reference data

Assumed: obtainable via Clearinghouse or FSA.

If unavailable: ghost student detection drops to weak proxies.

Entity resolution

Assumed: no enterprise capability exists today.

If one exists: we consume it rather than build it.

Rules platform

Assumed: build on an existing decision engine.

If none: buy versus build becomes a committee decision.

Detection strategy by typology

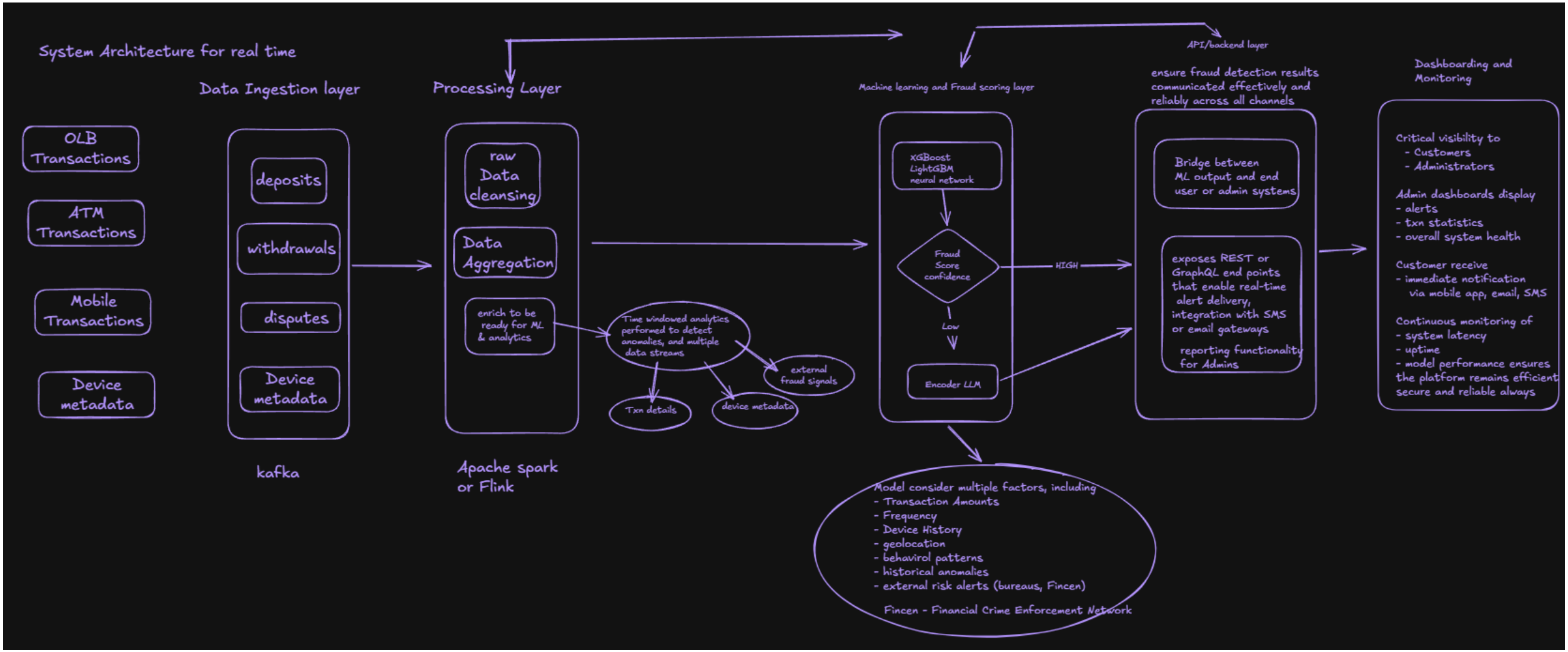
Different problems need different techniques, not one model

Typology	Primary signals	Technique	Key dependency
Ghost student	No enrolment record, thin file, SSN issuance mismatch, minor as recipient	Deterministic rules plus reference data joins	External enrolment data
Straw student	Shared device, address or contact across unrelated recipients, cluster formation	Entity resolution and graph clustering	Entity resolution capability
Money mule	Outflow velocity, cash and P2P share, dormancy then burst, new business payees	Supervised model, gradient boosting	Labelled outcomes and feedback loop

Sequencing: rules and reference joins ship first because the model risk governance clock is shorter, not because they detect better.

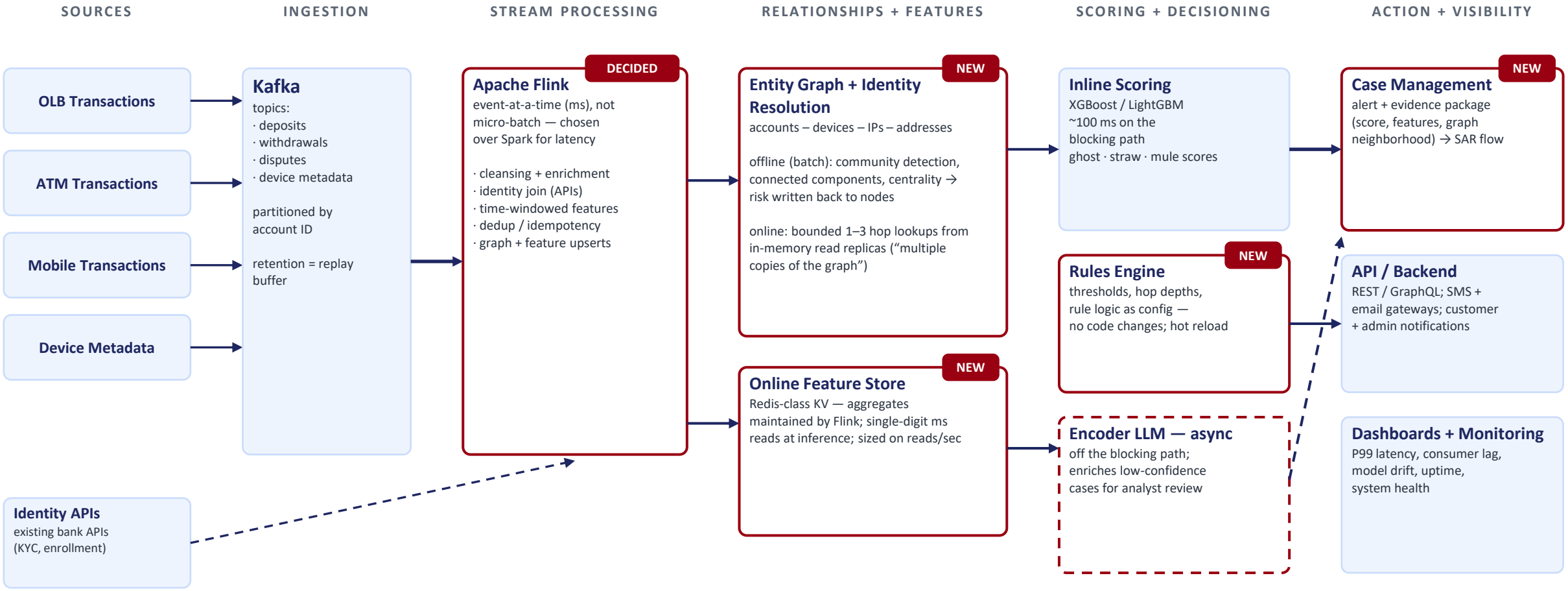
System Architecture Diagram: Fraud Detection

Trying to build a generic reference architecture for the common use cases



Real-Time Fraud Detection — Updated Architecture

Patched: Flink decided over Spark · identity resolution + entity graph · online feature store · LLM moved off the blocking path · case management added



Latency budget (P99 ≤ 500 ms): Kafka ingest ~50 ms → Flink processing ~100 ms → graph + feature lookups ~50 ms → model inference ~100 ms → headroom ~200 ms · LLM + batch graph analytics run off the clock

Scaling: every tier horizontal — Kafka partitions · Flink keyed state (RocksDB + checkpoints) · stateless model servers · graph read replicas · replicated KV feature store

we'd flag when a new account was within two hops of a known fraud cluster

How Graphs Deliver Real-Time Fraud Detection at Scale

FinCEN Student-Aid Fraud Alert · Ghost Students · Straw Students · Money Mules

Graph theory buys low latency.
Architecture buys scalability.

Why a Graph Fits Fraud

- Fraud is a network behavior — accounts share devices, IPs, addresses; refunds fan out to mules.
- Index-free adjacency: each node stores its neighbors, so query cost = neighborhood size, not database size.
- Risk scoring becomes a local computation — traverse outward from the event, never rescan the world.

Low Latency: Split Online / Offline

- Offline (batch): community detection, connected components, centrality — results written back as node risk properties.
- Online (per Kafka event): bounded 1–3 hop traversal + precomputed score lookups feed the ML model.
- Heavy graph theory runs ahead of time; the scoring path stays in milliseconds.

Scalability: Architecture

- Read replicas — "multiple copies of the graph": 35K nodes / 25M edges fits in memory; replicate and load-balance reads.
- Streaming upserts from Kafka mutate the graph incrementally — never rebuild.
- Depth caps + supernode handling bound worst-case query cost, making a latency SLA possible.

Same Graph, Three Detectors

Ghost Students

Connected components over shared PII, devices & IPs collapse hundreds of "different" applicants into one synthetic-identity cluster.

Straw Students

Community detection places real identities inside fraud-ring clusters; recruiter links appear as edges to known ring members.

Money Mules

Directed short-path traversal catches rapid refund → withdrawal hop chains right as the triggering event arrives.

Configurable without code changes: traversal patterns, hop depths, and risk thresholds live as parameterized queries & rules — satisfying the Principal Engineer's requirement.

System Requirements

Trying to build a generic reference architecture for the common use cases

Functional Requirements

- **Multi-factor authentication and secure login**
  The system must support multiple layers of authentication, including biometrics, OTPs, and device verification, ensuring only authorized users can access sensitive banking services.

- **Real-time monitoring of all transactions**
  Every transaction — deposits, withdrawals, fund transfers, or payments — must be captured instantly to enable timely analysis and detection of potential fraud.

- **Immediate alerting for high-risk activities**
  Suspicious or high-risk transactions must trigger alerts instantly, notifying administrators and customers through dashboards, mobile apps, email, or SMS, enabling prompt action.

- **Admin and customer dashboards with analytics**
  Dashboards should provide actionable insights and visual analytics. Administrators can review flagged transactions, while customers can monitor account activity efficiently.

- **Audit trail for regulatory compliance**
  The system must maintain comprehensive logs of all interactions, transactions, and alerts, ensuring transparency and enabling regulatory inspections and verification.

System Requirements

Trying to build a generic reference architecture for the common use cases

Non-Functional Requirements

- **Performance: Process transactions in milliseconds**
  The system must deliver high-speed processing to detect fraud in real-time, ensuring no delay compromises security.

- **Scalability: Handle spikes in transaction volume**
  It should accommodate sudden surges in transaction load, such as festive seasons or promotional campaigns, without affecting system responsiveness.

- **Availability: 24x7 uptime with failover and redundancy**
  Continuous system availability is critical, supported by failover mechanisms, redundant servers, and disaster recovery solutions.

- **Security: End-to-end encryption, tokenization, secure APIs**
  Comprehensive security must protect sensitive data at rest and in transit, using encryption, tokenization, and secure API connections.

- **Compliance: RBI, SEBI, AML, and KYC regulations**
  Regulatory compliance must be embedded in all processes, ensuring adherence to legal requirements and audit standards.

- **Usability: User-friendly dashboards and mobile app interface**
  Interfaces should be intuitive, enabling both administrators and customers to access insights, monitor activities, and act on alerts efficiently.

Risk Management Plan

Trying to build a generic reference architecture for the common use cases

- **Cybersecurity breach**
  The system faces high risk from potential cyberattacks. To mitigate this, end-to-end encryption, secure firewalls, regular penetration testing, and robust access controls are implemented to protect sensitive financial data from unauthorized access.

- **Model drift / accuracy decay**
  Machine learning models can lose accuracy over time due to changes in user behavior or new fraud patterns. Continuous retraining pipelines, monitoring performance metrics, and validating outputs against historical data ensure the models remain effective and reliable.

- **System downtime / outage**
  Any interruption in the system could result in delayed fraud detection and financial loss. Redundant servers, load balancing, and automatic failover mechanisms are incorporated to maintain 24x7 availability and minimize disruption.

- **Regulatory non-compliance**
  Failure to comply with RBI, SEBI, AML, and KYC regulations can result in penalties and reputational damage. Audit trails, logging, and automated compliance checks ensure all processes adhere to legal requirements.

- **Insider threats / fraud**
  Malicious activity by internal staff can compromise security. Role-based access controls, anomaly detection, logging, and regular audits help detect and prevent insider threats effectively.

- **User alert fatigue**
  Overloading customers with notifications may reduce responsiveness. Alerts are prioritized, made clear and actionable, and presented through user-friendly dashboards to maintain engagement without causing fatigue.

Business Impact & Strategic Benefits

Trying to build a generic reference architecture for the common use cases

- **Reduced Fraud Losses**
  By detecting suspicious transactions in real-time, the system prevents fraudulent activities before they escalate, significantly reducing financial losses for both the bank and its customers.
- **Enhanced Customer Trust and Retention**
  Real-time alerts and visible security measures enhance customer confidence. Knowing their accounts are monitored proactively increases loyalty and engagement with banking services.
- **Operational Efficiency**
  Automation of fraud detection processes reduces manual investigation workload, allowing staff to focus on high-priority cases while improving overall system responsiveness.
- **Regulatory Compliance and Audit Readiness**
  The system maintains comprehensive audit trails, log management, and explainable AI outputs. This ensures that all operations are compliant with banking regulations and ready for inspection at any time.
- **Scalability and Future Readiness**
  Designed to handle millions of transactions per day, the system can accommodate growth in user base, transaction volume, and evolving fraud techniques, making it adaptable to future banking needs.
- **Improved Decision-Making and Insights**
  Dashboards and analytics provide actionable insights for administrators, enabling data-driven decisions regarding risk mitigation, policy updates, and operational improvements.

Example Case: Pilot implementation in a mid-size bank reduced fraud alerts by **22% false positives** while catching **15% more genuine fraud cases** compared to legacy systems.


Design Lessons & Best Practices:
Fraud Detection System
Modular Architecture
Independently scalable components (ingestion, processing ML) allow easy upgrades without disruption
Explainable AI
SHAP & LIME tools provide transparency, reduce bias, and build in automated decisions
Data Versioning & Lineage
Tracking datsetts and feature versions ensurs auditabilty, auditabiity, reprousoubliity, and compliance
Latency vs. Cost Balance
Real-time scoring for high-risk; micro-batches for low-risk. Optimizes Optimizes performance and infrastrcture costs
Compliance by Design
Audit logs, trasieblity, and regulatory controls embedded to adhere fo RBI, SEBI, AML, KYC standards

Conclusion

Trying to build a generic reference architecture for the common use cases

A modern real-time fraud detection system goes beyond technology — it is the backbone of secure, customer-centric banking. By integrating big data pipelines, AI/ML models, and intuitive dashboards, banks can detect and respond to fraudulent activities instantly, minimizing financial losses and operational disruptions.

Such systems also ensure regulatory compliance with standards like RBI, SEBI, AML, and KYC, thanks to embedded audit trails and traceable data flows. Beyond compliance and security, real-time fraud detection enhances customer trust, showing users that their assets are monitored proactively and securely.

Moreover, the modular, scalable, and explainable design allows banks to adapt to evolving threats, upgrade components without downtime, and maintain cost-efficient operations. In an increasingly digital financial landscape, implementing a robust fraud detection platform is not just a technical necessity — it is a strategic advantage that strengthens reputation, operational efficiency, and long-term customer loyalty.

Question

- What does "near real time" mean—100 ms, 500 ms, or a few seconds?
- Is the risk score advisory, or can it place a hold on funds?
- Can we return an ACH credit, or only generate an alert?
- What enrollment/reference data is available and what is its SLA?
- How much historical labeled fraud data exists?
- Does the bank already have an enterprise entity-resolution/graph platform?
- What transaction volume and peak TPS are we designing for?
- What is the acceptable false-positive rate?
- Who owns rule approval and production promotion?
- What regulatory/model-risk constraints apply to the ML model?

Additional material

Latency questions

- **"What's your end-to-end latency budget, and how is it allocated?"** They want a number — e.g., "risk score within 500ms of the ACH credit landing on Kafka" — and a breakdown: ingestion ~50ms, stream processing ~100ms, feature lookup ~50ms, model inference ~100ms, etc. If you can't decompose the budget, "near real time" sounds hand-wavy.
- **"Spark or Flink — which, and why?"** ⚠️ Your slide says "Spark or Flink," and they will make you pick. Spark Structured Streaming is micro-batch (latency in seconds); Flink is true event-at-a-time streaming (milliseconds). For near-real-time scoring, Flink is the defensible answer — know that trade-off cold.
- **"What's the latency of the Encoder LLM path?"** ⚠️ This is the biggest target on your diagram. LLM inference is hundreds of ms to seconds — an order of magnitude slower than XGBoost. Be ready to say the LLM path is async/second-tier (enrichment for analyst review, not inline blocking of the transaction), or they'll flag it as breaking your real-time SLA.
- **"Where do features come from at inference time?"** Computing features on the fly kills latency. The expected answer is a low-latency feature store (Redis/online store) with precomputed aggregates, and time-windowed features maintained by the stream processor.
- **"What happens to latency when a dependency is slow?"** Identity API times out, feature store misses — do you block the transaction, score with degraded features, or fail open/closed? Fraud systems have a real answer here: score with what you have, flag the gap.

•**"Is scoring synchronous or asynchronous with the transaction?"** Do you block the ACH posting on the risk score, or score alongside and claw back? Big design fork; know your position.
•**"P50 vs P99?"** Averages hide tail latency. Committees love asking how you handle the 99th percentile — the answer involves bounded work per event, timeouts, and circuit breakers.

Scalability questions

•**"What throughput are you designing for, and what's the headroom?"** Events/sec now, at peak (refund disbursement dates are bursty — financial aid drops in waves at semester start), and at 10x. Kafka partitioning strategy: partitioned by what key? Account ID? What about hot partitions?
•**"How does the ML scoring layer scale?"** Stateless model servers behind a load balancer scale horizontally — easy answer, but say it. Then the harder follow-up: does the *feature store* scale with it? Reads per score × scores per second.
•**"How do you scale state in the stream processor?"** Time-windowed aggregations hold state. Flink checkpointing, state backend sizing, and what happens on rebalance/recovery — this is where senior candidates separate from mid-level.
•**"What happens during a consumer lag spike or replay?"** If you fall behind on Kafka, do stale events still get scored? Is there a backpressure strategy? Can you replay a day of events without double-alerting?
•**"How do you scale the relationship/link analysis?"** ⚠️ Your diagram has no graph or entity-resolution component — yet the case's core signals (shared devices, addresses, IPs across accounts) are relationship signals, and this whole conversation's graph story is your answer. Expect: "How do you detect that 200 accounts share a device — at scale, in real time?" That's your cue for the graph layer: precomputed offline analytics, bounded k-hop lookups online, read replicas for scale.

Cross-cutting traps to prep for

- **"Model retraining and drift — how do new fraud patterns get in without downtime?"** Shadow deployment, champion/challenger, feature backfill.
- **"How does 'configurable without code changes' work without adding latency?"** Rules engine / parameterized thresholds evaluated in-stream — and how a config change propagates (hot reload vs. redeploy).
- **"What's your fallback if the ML service is down entirely?"** Rules-only degraded mode is the classic answer.
- **"Where's the case management handoff?"** A HIGH score fires an alert — to whom, into what queue, with what SLA? Fraud ops people on the committee will ask.

What Interviewers Are Evaluating

In fraud detection system design interviews, the goal is not to test your knowledge of specific algorithms but to understand how you approach complex systems. Interviewers want to see whether you can design a system that balances speed, accuracy, and scalability.
If your solution only focuses on detecting fraud without considering latency or user experience, it signals an incomplete understanding. On the other hand, when you naturally incorporate trade-offs and real-world constraints, your design becomes much more compelling.

Why This Problem Is Challenging

Fraud detection is inherently difficult because it involves uncertainty and constantly evolving patterns. Unlike deterministic systems, where outcomes are predictable, fraud detection relies on probabilities and risk assessment.

Challenge	Description	Impact On Design
Real-Time Decisions	Must evaluate transactions instantly	Requires low-latency systems
Evolving Fraud Patterns	Fraudsters adapt quickly	Requires continuous updates
Data Volume	High transaction throughput	Demands scalable architecture

From Simple Validation To Intelligent Systems

In simpler systems, validation might involve checking whether inputs meet certain criteria. Fraud detection goes far beyond this by analyzing behavior, history, and patterns to make decisions.
When you design fraud detection systems, you move from static rules to dynamic decision-making. This shift is what makes the problem both interesting and valuable in interviews.

Understanding The Core Goals Of A Fraud Detection System

Before you start designing the architecture, you need to clearly understand what the system is trying to achieve. Many candidates jump straight into components without defining goals, which leads to unfocused designs.

Detecting Fraud Without Disrupting Users

The primary goal of a fraud detection system is to identify suspicious activity and prevent fraudulent transactions. However, this must be done without negatively impacting legitimate users.
If your system blocks too many genuine transactions, it creates friction and damages user experience. This balance between security and usability is a central challenge in fraud detection. → minimize reputation risk

Minimizing False Positives And False Negatives

A key aspect of fraud detection is managing errors, which are categorized as false positives and false negatives. False positives occur when legitimate transactions are flagged as fraud, while false negatives occur when fraudulent transactions are missed.
Designing a system that minimizes both types of errors requires careful consideration of thresholds and decision logic.

Error Type	Meaning	Business Impact
False Positive	Legitimate activity flagged as fraud	Poor user experience
False Negative	Fraud not detected	Financial loss

Supporting Investigation And Feedback

Fraud detection systems are not standalone solutions, as they often involve human analysts who review flagged cases. This means your system should support investigation workflows and provide relevant data for decision-making.
Feedback from these investigations is also critical because it helps improve the system over time. This creates a feedback loop that enhances both rules and models.

Balancing Accuracy And Performance

Another important goal is balancing accuracy with system performance. High accuracy often requires more complex analysis, which can increase latency and cost.
When you design your system, you need to decide how much complexity is acceptable for real-time decision-making. This trade-off is a key discussion point in interviews.

Key Components Of A Fraud Detection Architecture

Once you understand the goals, the next step is breaking the system into components. A well-structured architecture makes it easier to explain your design and ensures that all aspects of the system are covered.

Thinking In Layers And Components

Fraud detection systems are typically composed of multiple layers that work together to process data and make decisions. Each layer has a specific role, and understanding these roles helps you design a cohesive system.

When you organize your system into components, you create clarity in both design and communication. This is especially important in interviews where structure matters.

Core Architectural Components

A typical fraud detection system includes several key components that handle different stages of the workflow.

Component	Role	Example Function
Event Ingestion	Captures incoming data	Transaction or login events
Feature Store	Stores historical data	User behavior history
Rules Engine	Applies deterministic logic	Threshold checks
ML Scoring Service	Evaluates risk	Fraud probability scoring
Decision Engine	Determines action	Approve or block transaction

Interview Insight: Balancing Security And Experience

In interviews, decisioning is where you demonstrate product thinking. When you explain how your system balances fraud prevention with user experience, you show that you are not just building a technical solution but a practical one.

This ability to connect technical decisions with user impact is highly valued.

Scaling Fraud Detection Systems In Production

As your system moves from prototype to production, scaling becomes a critical concern. Fraud detection systems must handle large volumes of transactions while maintaining low latency and high accuracy.

Why Scaling Is Challenging In Fraud Detection

Unlike traditional systems, fraud detection requires real-time analysis for every transaction. This means your system must process high volumes of data without delays.

At the same time, you need to maintain accuracy and consistency, which adds complexity to scaling decisions.

Scaling The Ingestion And Processing Layers

The ingestion layer must handle a continuous stream of events, which requires horizontal scaling and efficient load distribution. Stream processing systems are often used to ensure that data is processed in real time.

Component	Scaling Approach	Benefit
Ingestion Layer	Partitioned streams	Handles high throughput
Feature Store	Distributed storage	Fast data access
Scoring Service	Stateless scaling	Low latency

Each component must scale independently to ensure overall system performance.

Handling Peak Traffic And Spikes

Fraud detection systems often experience traffic spikes during events like sales or peak business hours. Your system must be able to handle these spikes without degrading performance.
Auto-scaling mechanisms and buffering strategies help manage these fluctuations.
Designing for peak traffic ensures that your system remains reliable under stress.

Multi-Region Deployment And Availability

For global systems, deploying across multiple regions improves both latency and availability. This ensures that users receive fast responses regardless of their location.
However, multi-region deployments introduce challenges such as data consistency and increased costs. Being able to explain these trade-offs is important in interviews.

Interview Insight: Designing For Scale

In interviews, scaling discussions are often where candidates demonstrate their depth of understanding. When you explain how your system handles growth and high traffic, you show that you can design systems for real-world conditions.
This is a key indicator of strong system design skills.

Monitoring, Feedback Loops, And Model Improvement
A fraud detection system is never static, as fraud patterns evolve continuously. This makes monitoring and feedback essential for maintaining system effectiveness over time.

Why Continuous Monitoring Is Necessary
Without monitoring, you have no way of knowing whether your system is performing correctly. Metrics such as detection accuracy and latency provide insights into system behavior.
Monitoring allows you to identify issues early and make adjustments before they impact users or business outcomes.

Key Metrics For Fraud Detection Systems To evaluate performance, you need to track specific metrics that reflect both accuracy and efficiency.

Metric	What It Measures	Importance
Precision	Correct fraud detections	Reduces false positives
Recall	Fraud cases detected	Prevents missed fraud
Latency	Decision time	Ensures real-time response
False Positive Rate	Incorrect flags	Impacts user experience

Building Feedback Loops

Feedback loops are critical for improving both rules and machine learning models. Data from human reviews and confirmed fraud cases can be used to refine detection logic.
This continuous learning process ensures that your system adapts to new fraud patterns. It also improves accuracy over time.

Handling Model Drift And Changing Patterns

Fraud patterns change as attackers adapt to detection methods. This can lead to model drift, where the model's performance degrades over time.
Regular retraining and updating of models help address this issue. Including this in your design shows that you understand long-term system maintenance.

Interview Insight: Designing For Evolution

In interviews, discussing monitoring and feedback demonstrates that you are thinking beyond initial deployment. It shows that your system is designed to evolve and improve over time.
This forward-thinking approach is a strong signal of engineering maturity.

How To Answer Fraud Detection System Design Questions In Interviews

Understanding the system is only part of the challenge, as you also need to communicate your design effectively. Structuring your answer clearly is key to making a strong impression.

Starting With Requirements And Constraints

A strong answer begins with clarifying the problem and understanding requirements. This includes identifying the type of fraud, expected traffic, and latency constraints.
By starting with requirements, you ensure that your design is aligned with the problem. This also demonstrates a structured approach to problem-solving.

Defining The System Architecture

Once you understand the requirements, you should outline the main components of your system. This includes ingestion, feature computation, rules, machine learning, and decisioning layers.
Explaining how these components interact helps the interviewer follow your thought process. It also ensures that your design is complete.

Incorporating Optimization And Scaling

After presenting the basic design, you should discuss how your system handles **scaling and optimization**. This includes strategies for managing high traffic and maintaining low latency.
This step shows that you are thinking beyond the initial design and considering real-world challenges.

Explaining Trade-Offs Clearly

One of the most important parts of your answer is explaining trade-offs. This includes balancing speed, accuracy, and user experience.
When you articulate these trade-offs clearly, you demonstrate strong decision-making skills. This is often what distinguishes top candidates.

Interview Insight: Structured Thinking Wins

In interviews, your goal is to show how you think rather than just what you know. A clear and structured approach makes your answers more compelling.
When you guide the interviewer through your design logically, you create a strong impression.

Using structured prep resources effectively

Use Grokking System Design Interview: Patterns & Mock Interviews on Educative to learn curated patterns and practice full System Design problems step by step. It's one of the most effective resources for building repeatable System Design intuition. You can also choose the best System Design study material based on your experience:

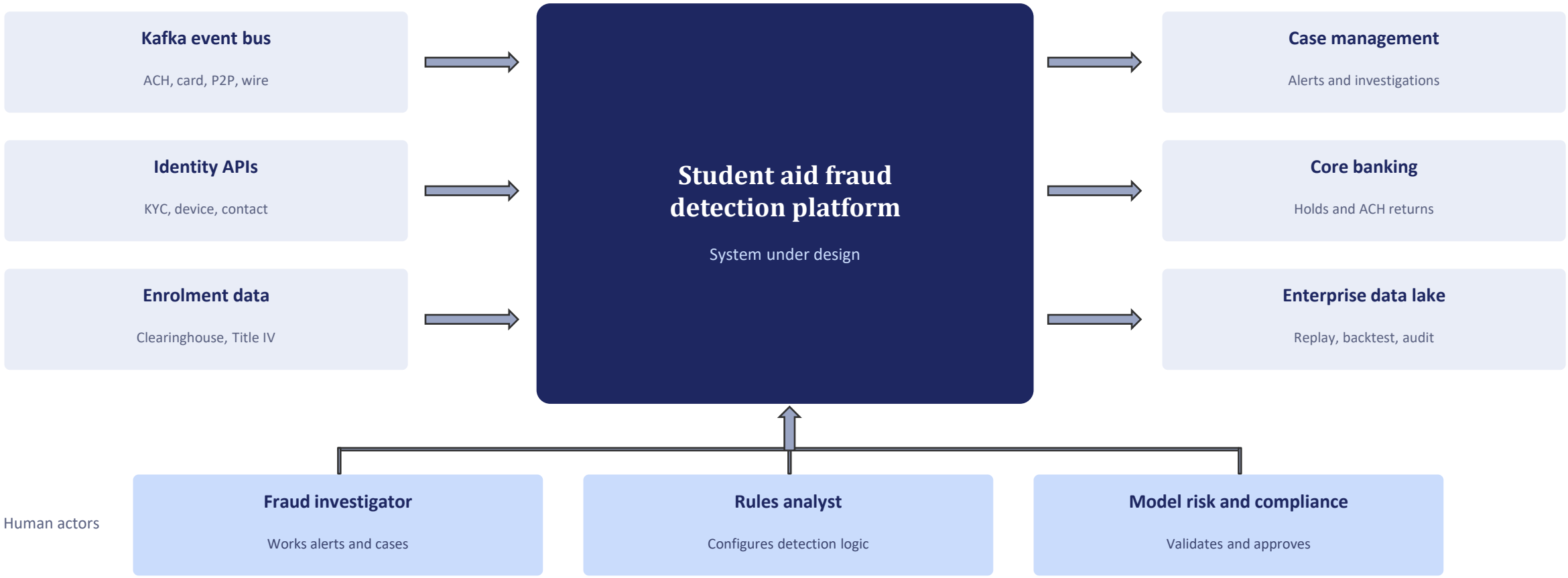
- 10 Best System Design Certification Options for Tech Career 2026

- 10 Best System Design Courses to Crack the Interview [2026]

- Top 10 platforms for System Design interviews

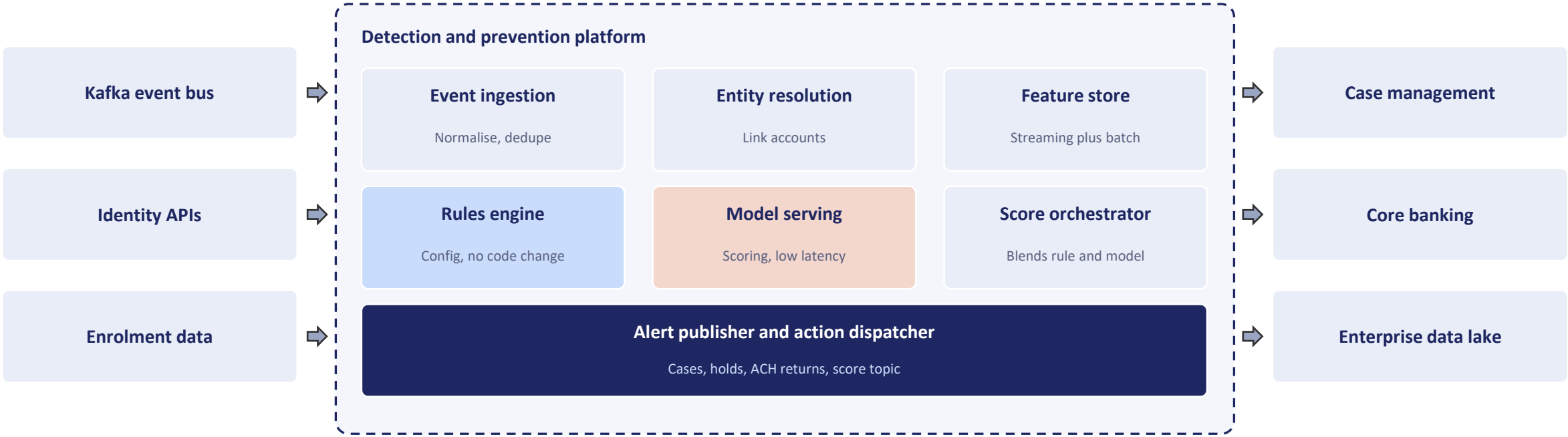
Context

System boundary and external dependencies



Target architecture

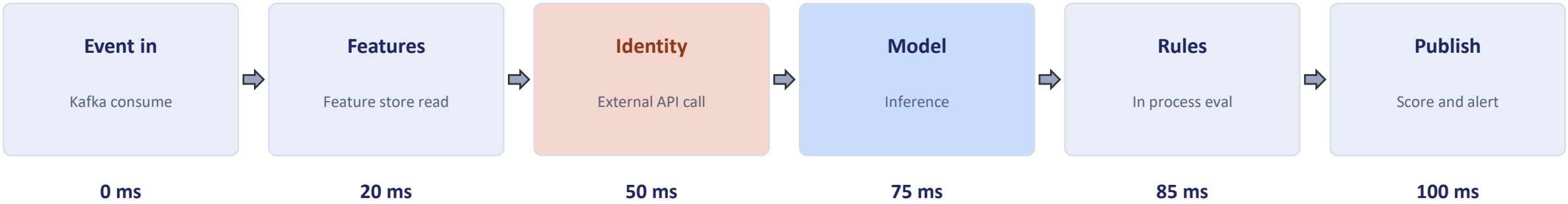
Hot path for scoring, analytical path for training, registry as the only coupling



Training never touches the hot path. The registry is the only coupling, and it is a pull of a versioned artefact.

Near real time path

Illustrative budget — replace with measured figures once the identity API SLA is confirmed



Riskiest dependency

The identity API is the only synchronous external call and is owned by another team. If it degrades, scoring degrades with it.

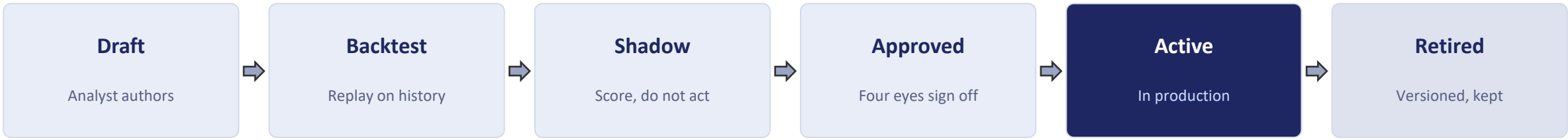
Preferred: hybrid. Cached attributes by default, synchronous call only for new accounts or cold cache — concentrating the expensive call where freshness changes the answer.

Two speed by design

- Per event scoring runs inline in the budget above.
- Network and cluster scoring cannot complete in 100ms — it runs on micro batch and updates an entity risk score.
- The hot path then reads that entity score as a feature.

Rule lifecycle

How "configurable without code changes" stays safe and auditable



1 Every transition is logged

Who changed what, when, and which version was live at the time of any given score.

2 No promotion without backtest

A rule cannot reach shadow until it has been replayed against history from the lake.

3 Rollback is a state change

Reverting is promoting a prior version, not editing production configuration.

This is what turns the configurability requirement from an audit finding into an engineering control.

Workflow to review readiness

The process to complete the design, not the runtime data flow

1	**Align on scope**	Stakeholder questions, success criteria, MVP typology	*Scope statement, decision log, assumptions register*
2	**Assess current state**	Topic inventory, identity API contracts, existing AML controls	*Gap analysis, build versus buy assessment*
3	**Design target state**	Architecture, detection strategy, ML approach, rule governance	*Diagrams, NFR mapping, lifecycle model*
4	**Validate with governance**	Model risk, compliance, legal, security, privacy pre-reads	*Signed pre-reads, threat model, regulatory position*
5	**Package for committee**	Trade-off analysis, roadmap, dependency map	*ARC submission pack*

Phases 2 and 3 iterate. What is sequential is the gating: we do not present to committee before governance pre-reads are complete.

Key decisions and next best alternatives

Every choice here is defensible either way — these are the trade-offs I made

Decision	Next best alternative	Why this way
Rules and reference joins ship before models	Single supervised model across all three typologies	Labels are sparse with long confirmation lag; governance clock on rules is shorter
Model serving as a separate component	Model embedded in the stream processor	Decouples model and application lifecycles; costs one network hop
Model score consumed as a rule input	Rules as a pre-filter, model on survivors	Lets analysts compose model and rule logic without code; costs more inference
Hybrid identity lookup, cached with selective call	Fully synchronous call on every event	Removes an external availability dependency from the hot path for most events

Delivery sequencing

Ordered by governance lead time and data dependency, not by technical appeal

Phase 1

Mule detection

Rules and velocity features on existing Kafka data. No external dependency, shortest governance path, immediate loss avoidance.

Phase 2

Straw student

Entity resolution and graph clustering across device, address and contact. Depends on ER capability decision.

Phase 3

Ghost student

Enrolment verification joins plus supervised scoring as labels mature. Depends on external data agreement.

Cross cutting from day one: the analytical path, rule governance, and audit retention of scores and features. These are controls, not phase two features — a regulator asking why an account was not flagged in March needs the score as it existed in March.

Where would you like to go deeper?

Backup

UML sequence diagram with timeout and alt branches

Backup

Kafka topic inventory and data contract questions

Backup

Model risk, fairness testing and reason code approach

Backup

Regulatory position on holds, returns and Reg CC

Open questions and assumptions are documented in the register on slide 3.